

DECLARATION

We, **ANANYA M** bearing USN: 01JST20IS008, **DHANYA VINAYAK HEGDE** bearing USN: 01JST20IS016, and **DHANIKA GOWDA D** bearing USN: 01JST20IS054 students of 8th semester of UG program, Information Science Branch, Department of Information Science and Engineering, JSS Science and Technological University, Mysuru hereby declare that the project work entitled “**DESIGN OF SPELL-CHECKER AND GRAMMAR-CHECKER FOR KANNADA DOCUMENTS**” has been carried out by us. This project is submitted to **JSS Science and Technological University**, Mysuru in partial fulfillment of the course requirements for the award of the Degree in **INFORMATION SCIENCE AND ENGINEERING** during the academic year 2023-24. This written submission represents a record of original work and we have adequately cited and referenced the original sources.

Further the matter embodied in this thesis has not been submitted to any other University or Institution for the award of any degree.

Place: Mysuru

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ABSTRACT

This project introduces a comprehensive Grammar Error Correction (GEC) tool designed specifically for the Kannada language. The tool integrates both rule-based and machine learning techniques to effectively address spelling and grammatical errors in Kannada text. For spell checking and correction, the Kannada spell-checker underwent testing on a sample of 20 words with errors, providing expected corrections for a majority of the words. The tool leverages the capabilities of the Hunspell library, offering features such as morphological analysis and affix file support, making it well-suited for handling Kannada language intricacies. In the realm of grammatical error detection, the Kannada model is trained using the DistilBERT pre-trained model, demonstrating effectiveness in identifying grammatical errors in Kannada text. Furthermore, the project includes a rule-based approach for grammatical error correction. This module was evaluated on 20 randomly selected Kannada sentences in present and future tenses, providing expected corrections for a majority of the sentences. Overall, the developed GEC tool showcases promising results across its components, indicating its potential to significantly enhance the quality of written Kannada. With its robust performance and versatility, the tool contributes valuable solutions to the challenges of Kannada language processing and demonstrates the synergy between traditional linguistic approaches and modern machine learning techniques in the field of Natural Language Processing (NLP).

ACKNOWLEDGEMENT

An endeavor is successful only when it is carried out under proper guidance and blessings. We would like to thank a few people who helped us in carrying out this work by lending invaluable assistance.

We express our sincere regards to His Holiness **Jagadguru Sri Shivaratri Deshikendra Mahaswamiji** who has showered his blessings on us for framing our career successfully. We hereby thank **Dr. C Nataraju**, Principal, SJCE, JSSTU, Mysuru and **Dr. S P Shiva Prakash**, HOD, Department of Information Science and Engineering, SJCE, JSSTU, Mysuru who encouraged us at this venture.

It is our foremost duty to thank our project supervisor **Dr. R J Prathibha** for her encouragement, effective guidance, and valuable suggestions right from the beginning of this project till its completion.

We thank panel members for their support and guidance throughout the project. We also extend our regards to all the **teaching and non-teaching staff** of the Department of Information Science and Engineering for their direct or indirect support towards the completion of this project. We would also like to thank our family and friends for their constant support.

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